

Unit 3 Study Material – Programming with Python

Topics:

1. Lists
2. Tuples
3. Dictionaries
4. Functions
5. Recursive Functions

1. Lists in Python

A list is a collection of items stored in a single variable. Lists are ordered and mutable.

Example:

```
mylist = [10, 20, 30, 40]
print(mylist)
```

Common Operations:

- Access: mylist[0]
- Replace: mylist[1] = 50
- Insert: mylist.insert(2, 25)
- Remove: mylist.remove(30)
- Append: mylist.append(100)
- Search: 20 in mylist
- Sort: mylist.sort()

2. Tuples in Python

A tuple is an ordered collection that is immutable.

Example:

```
t = (1, 2, 3)
print(t[0])
```

Methods:

```
t.count(2)
t.index(3)
```

3. Dictionaries in Python

Dictionary stores data in key-value pair form.

Example:

```
student = {"name": "Saksham", "age": 19, "course": "CSE"}
```

Operations:

```
student["city"] = "Jaipur"
student["age"] = 20
del student["course"]
```

Traversal:

```
for key, value in student.items():
    print(key, value)
```

4. Functions in Python

A function is a block of code that performs a specific task.

Example:

```
def greet():
    print("Hello")
```

5. Arguments and Return

```
def add(a,b):
    print(a+b)
```

```
def square(n):
    return n*n
```

6. Recursive Function

```
def fact(n):
    if n==1:
        return 1
    else:
        return n * fact(n-1)
```

```
print(fact(5))    # 120
```

Important Questions:

- What is list?
- Difference between list and tuple.
- Explain dictionary operations.
- Explain recursion with example.